Defining and quantifying motion

STUDENT WORKSHEET

Name: _____ Class: _____ Date: _____

Multiple-choice questions — choose one answer

1. A car travels 120 m in 10 s. Its average speed is: A 10 m/s B 12 m/s C 1200 m/s D 0.083 m/s

2. A car moves from x = +6 m to x = −4 m. Its displacement is: A +10 m B +2 m C −10 m D −2 m

3. A car travels 20 m east and then 20 m west, in 10 s. Its average velocity is: A 4 m/s east B 4 m/s west C 0 m/s D 2 m/s east

4. A car changes velocity from +2 to +10 m/s in 4 s. Its average acceleration is: A +2 m/s^2 B +3 m/s^2 C +8 m/s^2 D +12 m/s^2

5. A car moves to the right but its velocity decreases. The acceleration is: A necessarily positive B zero C to the left D to the right

6. Which is a vector? A distance B average speed C displacement D time

7. A car has v_i = 2 m/s and v_f = 6 m/s over 2 s at constant acceleration. Its displacement is: A 4 m B 6 m C 8 m D 12 m

8. A car starts and finishes at the same point. Which statement must be true? A zero distance B zero displacement C zero speed throughout D zero acceleration throughout

Free-response: show all working and SI units

9. Journey 1: Calculate the average speed. State whether this determines velocity without further information.

10. Journey 2: Calculate displacement and average velocity; interpret the minus sign.

11. Journeys 3 and 4: Calculate acceleration, average velocity and displacement for each. Compare the two journeys.

12. Journey 5: Find acceleration and displacement, assuming uniform acceleration. Explain why the car is slowing down.

13. Journey 6: Determine the time when the car stops, its final velocity and its net displacement over 4 s. Calculate the total distance travelled.

14. A car travels 40 m east followed by 10 m west in 10 s. Determine (a) distance (b) displacement (c) average speed (d) average velocity.

15. Extension: A car begins at +3 m/s and accelerates at +2 m/s^2 for 5 s. Use $\Delta x = v_i \Delta t + \frac{1}{2}a(\Delta t)^2$ to calculate its displacement.

16. Extension: A car enters a straight stretch at 4 m/s and accelerates uniformly to 10 m/s over 21 m. Use $v_f^2 = v_i^2 + 2a\Delta x$ to find a.

Reflection

17. Explain in one sentence why a negative acceleration does not always mean a vehicle slows down.